

Disturbance rejection and convex optimization-based guidance for autonomous shipborne landing of a reusable rocket

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ABSTRACT

Landing a reusable rocket on a drone ship poses unique challenges as the ship's motion creates a significant disturbance not addressed in most rocket landing studies. To address this disturbance and achieve an optimal landing, this paper develops a dual-loop guidance framework that integrates disturbance rejection with convex optimization. The outer loop optimizes landing trajectories toward predicted deck states in real time. To rapidly optimize landing trajectories, a Hessian-enhanced sequential convex programming method is developed. It adaptively tunes penalty terms based on dynamics nonlinearity to accelerate convergence. The generated trajectory is then tracked by the inner loop, which features an internal model-based observer coupled with a deck oscillation frequency estimator. The system achieves exponential convergence despite unknown deck oscillation amplitudes and frequencies. Simulation studies validate the proposed guidance law, demonstrating robust convergence and high landing accuracy under significant drone ship oscillations.

1. Introduction

Shipborne landings present unique guidance challenges across aerospace applications. While shipborne landing guidance methods for fixed-wing aircraft [1], helicopters [2], and multirotors [3,4] have been proposed before, rocket landing on a drone ship presents a different control problem. Unlike aircraft that rely on aerodynamic lift to maintain forward velocity or possess hovering capabilities, vertical rocket landing requires propulsive deceleration to achieve a zero-velocity, precision touchdown under significant actuator constraints and significant disturbances. This demands the development of high-performance guidance laws that go beyond existing powered descent guidance (PDG) methods to achieve both accuracy and optimality.

Current research on PDG methods can be classified into three categories: analytical guidance, optimization-based guidance, and learning-based guidance. A more detailed classification is illustrated in Fig. 1. Polynomial guidance methods trace their origins to the Apollo Lunar descent guidance [5,6], which employs position and velocity feedback to track polynomial reference trajectories. Modern extensions include augmented polynomial forms [7] and fractional-order generalizations [8] that introduce additional parameters for trajectory shaping. Another analytical approach, zero-effort-miss/zero-effort-velocity (ZEM/ZEV) guidance, derives from acceleration-optimal control theory, where ZEM and ZEV quantify the terminal errors without further control

[9]. Various enhancements have been proposed to prevent ground penetration, including altitude penalties [10], waypoint optimization [11], and non-concave trajectory construction [12]. Sliding-surface guidance represents a third analytical approach, reformulating PDG as a position stabilization problem with sliding surfaces designed for finite-time convergence [13] and for monotonic descent [14]. These analytical guidance methods provide computationally efficient solutions for fixed landing targets.

Optimization-based guidance formulates PDG as an optimal control problem solved in real-time through indirect or direct methods. Indirect methods solve the necessary optimality conditions [15,16], while direct methods discretize the problem into nonlinear programming formulations [17–19]. Reference [20] demonstrates that the PDG problem can be convexified into second-order cone programming with theoretical convergence guarantees and efficient numerical solvers. This framework has been extended to incorporate six-degree-of-freedom dynamics [18,21,22], atmospheric effects [17,23–25], and parametric uncertainties [26]. When implemented in a receding-horizon framework, these methods create closed-loop systems that demonstrate resilience to various disturbances [21,23]. The reliability of these closed-loop solutions relies on the convergence properties of the underlying Sequential Convex Programming (SCP) algorithms, prompting recent innovations in algorithmic robustness. To mitigate artificial infeasibility, smoothing-homotopy techniques have been developed, utilizing approaches rang-

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Nomenclature			
Acronyms			
DOFE	Deck Oscillation Frequency Estimator	A_i, ω_i, ϕ_i	Amplitude, frequency, and phase of the i -th deck oscillation mode
DR-COGB	Disturbance Rejection and Convex Optimization-Based Guidance	C_B	Aerodynamic coefficient vector
HSCP	Hessian-enhanced Sequential Convex Programming	e_{ξ}	internal model-based observer estimation error
IMBO	Internal Model-Based Observer	e_r, e_v	Position and velocity tracking errors
Variables and Symbols		f	Dynamics function of the rocket
B	Body-fixed reference frame	g_I	Gravitational acceleration vector
ϵ_{tol}	Convergence tolerance for HSCP	n_B	Thrust direction vector in body frame
η, Ω	Augmented state and control variables	$r_{\mathcal{E}}, V_{\mathcal{E}}$	Relative position and velocity (rocket minus deck)
I	Inertial reference frame	r_I, V_I	Rocket position and velocity vectors in the inertial frame
$\mathbb{S}_+^{15}$	Cone of 15×15 symmetric positive semidefinite matrices	$r_{I,deck}, V_{I,deck}$	Drone ship deck position and velocity vectors
D, G	Observer design matrices	w	State vector of the drone ship oscillation model
$F_{\theta\theta}$	Hessian of dynamics with respect to trajectory	x, u	State vector and control vector
I_3, I_{6s}	Identity matrices of dimension 3 and 6	V_{ex}	Rocket engine exhaust velocity
K_{ϕ}	Attitude feedback gain matrix	F, F_{min}, F_{max}	Thrust magnitude and its lower/upper bounds
K_r, K_v	Feedback gain matrices for position and velocity	i_{max}	Maximum iteration limit for HSCP
L, S	Output matrix and system matrix of the deck oscillation model	k	Guidance cycle index
M	Solution matrix to the Sylvester equation	k_F, k_{ω}	Penalty weights for thrust magnitude and attitude rate
N_1, N_2	Deck oscillation frequency parameters	m, m_0	Rocket mass and initial mass
$Q_r, Q_v, Q_{\phi}, Q_m, Q_F$	Jacobian matrices of dynamics	N	Number of discretization mesh points
R_y, R_z	Elementary rotation matrices	n	Number of previous guidance cycles for DOFE
R_{IB}	Rotation matrix from body frame to inertial frame	s	Number of sinusoidal modes in deck oscillation
X_k, Y_k, Z_k	Aggregated matrices for DOFE regression	t, t_0, t_f	Physical time, initial time, and final landing time
$\Pi_{\mathbb{S}_+^{15}}$	Projection operator onto positive semidefinite cone	$t_{predict}$	Time threshold for deck prediction activation
ρ	Atmospheric density	t_{stop}	Time-to-go threshold for stopping trajectory updates
S_{ref}	Aerodynamic reference area	Subscripts and Superscripts	
τ	Normalized time variable, $\tau \in [0, 1]$	$(\cdot)_B, (\cdot)_I$	Expressed in Body-fixed frame or Inertial frame
λ	Costate vector	$(\cdot)^{[k]}$	Variable corresponding to the k -th guidance cycle
ω	Attitude rate vector	$(\cdot)^*$	Optimal value
ϕ	Attitude Euler angles (yaw, pitch)	$(\cdot)^{\theta}$	Reference value drawn from the trajectory guess
θ	Trajectory guess vector		
ξ	State vector of the internal model-based observer		
A_0	Mean position of the drone ship deck		
A_I	Aerodynamic force vector expressed in the inertial frame		

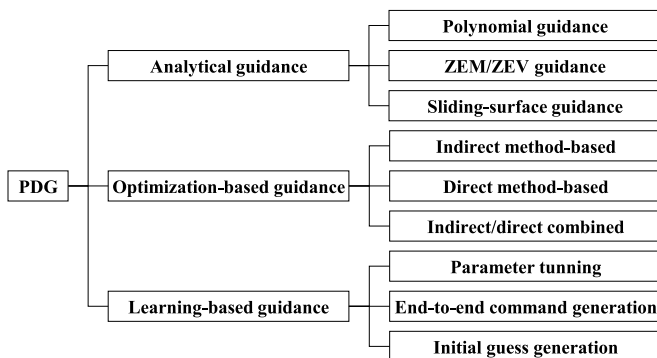


Fig. 1. Categorization of PDG guidance methods.

ing from terminal state convolution to full-trajectory NURBS parameterization [27,28]. Enhancing this stability, adaptive trust-region strategies now leverage differential algebra to tailor step sizes based on state-dependent nonlinearities [29]. Furthermore, to satisfy strict on-board timing requirements, computational efficiency has been significantly improved through reduced-space formulations that employ low-complexity matrix inversion schemes [30]. Nevertheless, a fundamental limitation persists in standard SCP formulations, which predominantly rely on first-order Taylor expansions to construct the convex subproblems. As a result, second-order curvature information remains largely unexploited, potentially limiting convergence rates in highly nonlinear scenarios. These optimization-based guidance methods provide propellant-efficient trajectories for fixed targets, but have yet to be extended to moving targets like drone ships.

Learning-based guidance employs neural networks to develop guidance policies from simulated data. Neural networks can serve multiple roles in PDG. Reference [31] integrates reinforcement learning with classical ZEM/ZEV guidance through an actor-critic framework that learns optimal guidance parameters. Reference [32] trains deep neural networks to assess the feasibility of propellant-optimal landing trajectories. Reference [33] uses neural networks to provide initial guesses for numerical optimization methods. Reference [34] presents a physics-informed neural network for parameter tuning a parameterized baseline guidance law for tackling stochastic uncertainties. While these data-driven approaches demonstrate computational efficiency and the ability to solve complex PDG problems, shipborne landings violate their fixed-terminal assumptions.

While traditional PDG methods are effective for static targets, they lack specific mechanisms to handle the disturbances introduced by sea-based platforms, which often leads to degraded landing precision and robustness. This study is motivated by the need to overcome this specific limitation. To handle disturbances caused by ship motion and achieve optimal landing, this paper presents a disturbance rejection and convex optimization-based guidance (DR-COBG) law. While similar optimal and disturbance rejection guidance frameworks have been explored in previous literature [24,25,34], none of them address disturbances in the landing target, which significantly complicate the problem. To overcome these challenges, this paper makes three key contributions:

1) To achieve optimal and accurate landing on a drone ship, a dual-loop guidance framework is developed. Unlike existing works that view drone ship motion as a disturbance, this framework incorporates drone ship motion predictions into trajectory optimization. The shipborne landing guidance problem is decomposed into an optimal guidance problem for predicted deck motion and a disturbance rejection guidance problem for unpredicted deck motion. The two subproblems are more tractable.

2) To ensure computational efficiency for real-time applications, the outer-loop guidance, for optimizing landing trajectories, employs a new Hessian-enhanced sequential convex programming (HSCP) method. Unlike existing SCP methods that use first-order approximations of nonlinear dynamics, HSCP further exploits the second-order derivative of dynamics and costate to adaptively tune penalty terms, providing reliable and fast convergence.

3) To achieve high accuracy landing, the inner-loop guidance, for tracking optimized trajectories, uses an internal-model based observer (IMBO) to estimate and compensate for the deck oscillation, achieving exponential convergence. A deck oscillation frequency estimator (DOFE) is developed, enabling prediction of future deck motion. This differs from existing disturbance rejection guidance methods that estimate current drone ship motion, but cannot predict future motion for accurate landing.

This paper is organized as follows. Section 2 formulates the guidance problem. Section 3 gives the main result of this paper, developing the guidance law. Section 4 presents the numerical study results that demonstrate the performance of the proposed guidance law.

2. Problem formulation

2.1. Optimal control formulation

The shipborne landing guidance problem is formulated as an optimal control problem. It uses two reference frames, an inertial reference frame I and a body-fixed reference frame B , as illustrated in Fig. 2. The state variables $(\mathbf{r}_I, \mathbf{V}_I, \phi, m)$ represent the reusable rocket's position, velocity, attitude, and mass, respectively. The attitude $\phi = (\phi_{\text{yaw}}, \phi_{\text{pitch}})$ is parameterized by yaw and pitch angles. The control variables (F, ω) represent the thrust magnitude and attitude rate, respectively. The optimal control problem is given as follows.

Problem 1. Shipborne landing guidance problem

$$\begin{aligned} & \text{2minimize} \quad \int_{t_0}^{t_f} k_F F^2 + k_\omega \|\omega\|_2^2 dt \\ & \text{subject to} \quad \dot{\mathbf{r}}_I = \mathbf{V}_I \\ & \quad \dot{\mathbf{V}}_I = \mathbf{g}_I + (F \mathbf{R}_{IB} \mathbf{n}_B + \mathbf{A}_I)/m \\ & \quad \dot{\phi} = \omega \\ & \quad \dot{m} = -F/V_{\text{ex}} \\ & \quad \dot{\mathbf{w}} = (\mathbf{S} \otimes \mathbf{I}_3) \mathbf{w} \\ & \quad F_{\min} \leq F \leq F_{\max} \\ & \quad \mathbf{r}_I(t_0) = \mathbf{r}_0, \mathbf{V}_I(t_0) = \mathbf{V}_0, \phi(t_0) = \phi_0, m(t_0) = m_0, \mathbf{w}(t_0) = \mathbf{w}_0 \\ & \quad \mathbf{r}_I(t_f) = \mathbf{r}_{I,\text{deck}}[\mathbf{w}(t_f)], \mathbf{V}_I(t_f) = \mathbf{V}_{I,\text{deck}}[\mathbf{w}(t_f)], \phi(t_f) = (0, \pi/2) \end{aligned}$$

In Problem 1, t_0 and t_f are the initial and final times. k_F and k_ω are the control penalties. $\mathbf{g}_I$ is gravitational acceleration. $\mathbf{R}_{IB}$ transforms from body to inertial frame. $\mathbf{n}_B$ is the thrust direction. $\mathbf{A}_I$ is the aerodynamic force. V_{ex} is the exhaust velocity. $\mathbf{S}$ is a matrix that formulates the drone ship motion. (A detailed definition is provided in the following subsection.) $F_{\min}$ and $F_{\max}$ are the minimum and maximum thrust magnitudes. The expressions of $\mathbf{R}_{IB}$ and $\mathbf{A}_I$ are given by

$$\mathbf{R}_{IB}(\phi) = [\mathbf{R}_y(\phi_{\text{yaw}}) \mathbf{R}_z(\phi_{\text{pitch}})]^T \quad (1)$$

$$\mathbf{A}_I(\mathbf{r}_I, \mathbf{V}_I, \phi) = \frac{1}{2} \rho(r_I) \|\mathbf{V}_I\|^2 S_{\text{ref}} \mathbf{R}_{IB}(\phi) \mathbf{C}_B [\mathbf{R}_{BI}(\phi) \mathbf{V}_I] \quad (2)$$

where $\mathbf{R}_y$ and $\mathbf{R}_z$ are elementary rotation matrices about the y and z axes. ρ is the density of air. S_{ref} is the reference area. $\mathbf{R}_{BI} = \mathbf{R}_{IB}^T$. $\mathbf{C}_B$ describes the aerodynamic coefficients.

2.2. Ship motion formulation

The distinction from conventional PDG problems lies in the terminal constraints that require the rocket to have the same position and velocity as the drone ship deck, denoted by $\mathbf{r}_{I,\text{deck}}$ and $\mathbf{V}_{I,\text{deck}}$, respectively. To model the deck oscillation due to sea conditions, a sum-of-sinusoids model from existing shipborne landing studies [1–4] is employed:

$$\mathbf{r}_{I,\text{deck}}(t) = \mathbf{A}_0 + \sum_{i=1}^s \mathbf{A}_i \sin(\omega_i t + \phi_i) \quad (3)$$

where $\mathbf{A}_0$ is the mean deck position. $\mathbf{A}_i$, ω_i , and ϕ_i are the amplitude, frequency, and phase of the i -th oscillation mode. The deck velocity $\mathbf{V}_{I,\text{deck}}$ is obtained by time differentiation of $\mathbf{r}_{I,\text{deck}}$.

Problem 1 introduces a state variable $\mathbf{w}$ to describe the motion of the drone ship, which satisfies the relations:

$$\mathbf{r}_{I,\text{deck}}(\mathbf{w}) = \mathbf{A}_0 + (\mathbf{L} \otimes \mathbf{I}_3) \mathbf{w} \quad (4)$$

$$\mathbf{V}_{I,\text{deck}}(\mathbf{w}) = (\mathbf{L} \mathbf{S} \otimes \mathbf{I}_3) \mathbf{w} \quad (5)$$

where $\mathbf{L} \in \mathbb{R}^{1 \times 2s}$ is an output matrix. $\mathbf{S} \in \mathbb{R}^{2s \times 2s}$ is a neutrally stable matrix that encodes the oscillation frequencies. $\mathbf{I}_3$ is the 3×3 identity matrix. “ $\otimes$ ” denotes the Kronecker product. The Kronecker product with $\mathbf{I}_3$ is for extending each oscillation mode to three-dimensional position and velocity components. The expressions of $\mathbf{S}$, $\mathbf{L}$, and $\mathbf{w}_0$ are given by

$$\mathbf{S} = \begin{bmatrix} 0 & -\omega_1 & & & & \\ \omega_1 & 0 & & & & \\ & & \ddots & & & \\ & & & \ddots & & \\ & & & & \ddots & \\ & & & & & 0 & -\omega_s \\ & & & & & \omega_s & 0 \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ \vdots \\ 0 \\ 1 \end{bmatrix}^T, \quad \mathbf{w}_0 = \begin{bmatrix} \mathbf{A}_1 \cos \phi_1 \\ \mathbf{A}_1 \sin \phi_1 \\ \vdots \\ \vdots \\ \mathbf{A}_s \cos \phi_s \\ \mathbf{A}_s \sin \phi_s \end{bmatrix} \quad (6)$$

It is assumed that current deck position $\mathbf{r}_{I,\text{deck}}$ and velocity $\mathbf{V}_{I,\text{deck}}$ are available, but the oscillation state $\mathbf{w}$ and the parameters $(\mathbf{S}, \mathbf{L}, \mathbf{w}_0, \mathbf{A}_0)$ are unavailable, which creates the main challenge in shipborne landing guidance because Problem 1 cannot be solved.

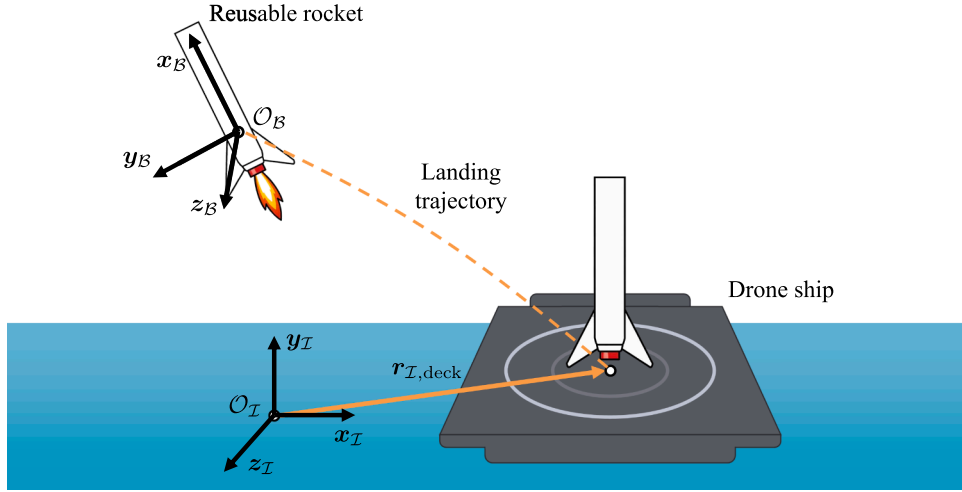


Fig. 2. Illustration of reference frames.

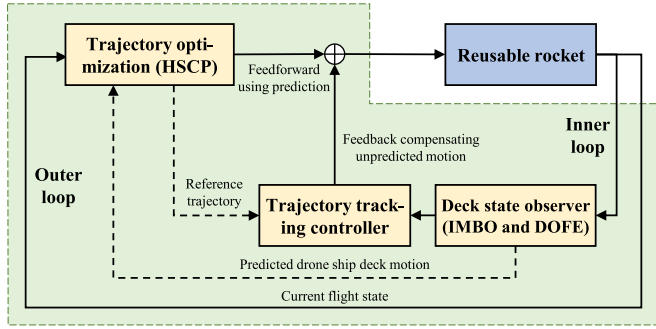


Fig. 3. Framework of the proposed DR-COBG.

3. Disturbance rejection and convex optimization-based guidance (DR-COBG)

3.1. Dual-loop guidance framework

To simultaneously reject disturbances caused by ship motion and achieve optimal landing performance, the proposed DR-COBG features a dual-loop framework, as illustrated in Fig. 3. The inner loop predicts drone ship motion and sends these predictions to the outer loop. The outer loop then uses these predictions to generate optimal landing trajectories, which are passed back to the inner loop. The inner loop tracks these optimized trajectories and rejects any unpredicted deck oscillations that weren't accounted for in trajectory generation, ensuring accuracy and optimal landing on drone ship decks.

3.2. Outer-loop guidance with HSCP

The outer-loop guidance generates optimal landing trajectories by solving optimal control problems in each guidance cycle with a novel HSCP method. Unlike the original problem formulation in Problem 1, which requires perfect knowledge of oscillation parameters, the outer-loop guidance uses the predicted drone ship motion from the inner loop to formulate a more tractable guidance problem with a virtual target. Solving this problem in real-time provides the reference trajectory that the inner loop tracks.

3.2.1. Shipborne landing guidance problem with virtual target

The outer-loop guidance solves the following optimal control problem in each guidance cycle:

Problem 2. Shipborne landing guidance problem with virtual target

$$\begin{aligned}
 &\text{minimize} && \eta(1) \\
 &\text{subject to} && \mathbf{r}'_I = (t_f - t_0)\mathbf{V}_I \\
 & && \mathbf{V}'_I = (t_f - t_0)[\mathbf{g}_I + (\mathbf{F}\mathbf{R}_{IB}\mathbf{n}_B + \mathbf{A}_I)/m] \\
 & && \boldsymbol{\phi}' = (t_f - t_0)\boldsymbol{\omega} \\
 & && m' = -(t_f - t_0)\mathbf{F}/V_{\text{ex}} \\
 & && \eta' = (t_f - t_0)\Omega \\
 & && F_{\min} \leq F \leq F_{\max}, \quad k_F F^2 + k_\omega \|\boldsymbol{\omega}\|_2^2 \leq \Omega \\
 & && \mathbf{r}_I(0) = \mathbf{r}_0, \mathbf{V}_I(0) = \mathbf{V}_0, \boldsymbol{\phi}(0) = \boldsymbol{\phi}_0, m(0) = m_0, \eta(0) = 0 \\
 & && \mathbf{r}_I(1) = \hat{\mathbf{r}}_{I,\text{deck}}^{[k]}(t_f), \mathbf{V}_I(1) = \hat{\mathbf{V}}_{I,\text{deck}}^{[k]}(t_f), \boldsymbol{\phi}(1) = (0, \pi/2)
 \end{aligned}$$

In Problem 2, a new state η and a control Ω are introduced to formulate the integral term in the performance index. There are three modifications from Problem 1 to Problem 2:

- 1) The deck position and velocity ($\mathbf{r}_{I,\text{deck}}, \mathbf{V}_{I,\text{deck}}$) are replaced by predicted values ($\hat{\mathbf{r}}_{I,\text{deck}}^{[k]}, \hat{\mathbf{V}}_{I,\text{deck}}^{[k]}$), where k is the guidance cycle number.
- 2) The time variable of states and controls is normalized by

$$t = t_0 + (t_f - t_0)\tau; \quad \tau \in [0, 1] \quad (7)$$

The original problem uses physical time $t \in [t_0, t_f]$, while the virtual problem uses normalized time $\tau \in [0, 1]$. And t_f becomes an optimizable parameter.

- 3) The integral term in performance index is reformulated using augmented state η and control Ω , transforming the integral term into a terminal term $\eta(1)$ with a convex constraint that becomes active at optimality.

3.2.2. Hessian-enhanced sequential convex programming (HSCP)

To solve Problem 2 in real-time, this paper develops a HSCP method that exploits second-order derivatives of dynamics and costate for reliable and fast convergence. Let θ and λ denote guess about the trajectory and the costate, such that

$$\theta(\tau) = [\mathbf{r}_I^\theta(\tau), \mathbf{V}_I^\theta(\tau), \boldsymbol{\phi}^\theta(\tau), m^\theta(\tau), \eta^\theta(\tau), F^\theta(\tau), \boldsymbol{\omega}^\theta(\tau), \Omega^\theta(\tau), t_f^\theta] \quad (8)$$

where the superscript θ denotes the variables are from a trajectory guess. HSCP approximates Problem 2 around (θ, λ) by a convex optimal control problem:

Problem 3. Convexified shipborne landing guidance problem with virtual target

$$\text{minimize} \quad \eta^x(1) + \frac{1}{2} \int_0^1 [(\mathbf{x}, \mathbf{u}, t_f) - \theta]^\top \Pi_{\mathbb{S}_+^5} [\lambda^\top \nabla_\theta^2 f(\theta)] [(\mathbf{x}, \mathbf{u}, t_f) - \theta] d\tau$$

$$\begin{aligned} \text{subject to } \mathbf{x}' &= \mathbf{f}(\theta) + \nabla_{\theta} \mathbf{f}(\theta)[(\mathbf{x}, \mathbf{u}, t_f) - \theta], \quad |(\mathbf{x}, \mathbf{u}, t_f) - \theta| \leq \kappa \\ F_{\min} &\leq F^u \leq F_{\max}, \quad k_F (F^u)^2 + k_{\omega} \|\omega^u\|_2^2 \leq \Omega^u \\ \mathbf{r}_I^x(0) &= \mathbf{r}_0, \mathbf{V}_I^x(0) = \mathbf{V}_0, \phi^x(0) = \phi_0, m^x(0) = m_0, \eta^x(0) = 0 \\ \mathbf{r}_I^x(1) &= \hat{\mathbf{r}}_{I, \text{deck}}^{[k]}(t_f^{\theta}), \mathbf{V}_I^x(1) = \hat{\mathbf{V}}_{I, \text{deck}}^{[k]}(t_f^{\theta}), \phi^x(1) = (0, \pi/2) \end{aligned}$$

In Problem 3, $\mathbf{x}$ is the state variable. $\mathbf{u}$ is the control variable. κ is the trust-region radius to ensure correctness of linearization. $\mathbf{f}$ is the time-normalized dynamical equation. $\nabla_{\theta} \mathbf{f}$ and $\nabla_{\theta}^2 \mathbf{f}$ are the Jacobian and Hessian of $\mathbf{f}$ with respect to θ . $\Pi_{\mathbb{S}_+^{15}}$ projects a matrix onto the cone of 15×15 symmetric positive semidefinite matrices. Their expressions are given by

$$\mathbf{x}(\tau) = [\mathbf{r}_I^x(\tau), \mathbf{V}_I^x(\tau), \phi^x(\tau), m^x(\tau), \eta^x(\tau)] \quad (9)$$

$$\mathbf{u}(\tau) = [F^u(\tau), \omega^u(\tau), \Omega^u(\tau)] \quad (10)$$

$$\mathbf{f}(\theta) = (t_f^{\theta} - t_0) \begin{bmatrix} \mathbf{g}_I + [F^{\theta} \mathbf{R}_{IB}(\phi^{\theta}) \mathbf{n}_B + \mathbf{A}_I(\mathbf{r}_I^{\theta}, \mathbf{V}_I^{\theta}, \phi^{\theta})]/m^{\theta} \\ \omega^{\theta} \\ -F^{\theta}/V_{\text{ex}} \\ \Omega^{\theta} \end{bmatrix} \quad (11)$$

$$\Pi_{\mathbb{S}_+^{15}}[\lambda^{\top} \nabla_{\theta}^2 \mathbf{f}(\theta)] = \arg\min_{\mathbf{H} \in \mathbb{S}_+^{15}} \|\mathbf{H} - \lambda^{\top} \nabla_{\theta}^2 \mathbf{f}(\theta)\|_F \quad (12)$$

where $\mathbb{S}_+^{15}$ is the cone of 15×15 symmetric positive semidefinite matrices. $\|\cdot\|_F$ is the Frobenius norm. HSCP evaluates the projection $\Pi_{\mathbb{S}_+^{15}}$ by thresholding negative eigenvalues in the eigendecomposition:

$$** \lambda^{\top} \nabla_{\theta}^2 \mathbf{f}(\theta) = \mathbf{Q} \text{diag}(\lambda_i) \mathbf{Q}^{\top}, \quad \Pi_{\mathbb{S}_+^{15}}[\lambda^{\top} \nabla_{\theta}^2 \mathbf{f}(\theta)] = \mathbf{Q} \text{diag}[\max(\lambda_i, 0)] \mathbf{Q}^{\top} \quad (13)$$

The relationship between Problems 1 to 3 is a solution transformation chain. The original guidance problem (Problem 1) is unsolvable due to its requirement for explicit ship oscillation parameters. By substituting a predicted virtual target, it transforms into Problem 2, which is then convexified to Problem 3. Upon convergence of the HSCP algorithm, the solution to Problem 3 satisfies the first-order necessary conditions of Problem 2 [35], thus effectively addressing the original guidance problem.

The HSCP method differs from standard SCP approaches through its incorporation of second-order information. HSCP augments the performance index with a quadratic penalty term:

$$J_{\text{HSCP}} = \frac{1}{2} \int_0^1 [(\mathbf{x}, \mathbf{u}, t_f) - \theta]^{\top} \Pi_{\mathbb{S}_+^{15}}[\lambda^{\top} \nabla_{\theta}^2 \mathbf{f}(\theta)] [(\mathbf{x}, \mathbf{u}, t_f) - \theta] d\tau \quad (14)$$

The novelty lies in three aspects: 1) the exploitation of second-order derivatives $\nabla_{\theta}^2 \mathbf{f}$ to capture nonlinear dynamics beyond first-order approximation, 2) the projection operator $\Pi_{\mathbb{S}_+^{15}}(\cdot)$ that maintains convexity by thresholding negative eigenvalues while preserving beneficial curvature information, and 3) the use of time-varying Hessian $\lambda^{\top} \nabla_{\theta}^2 \mathbf{f}(\theta)$ rather than a time-invariant penalty as in existing SCP works. The Hessian enhancement accelerates convergence as demonstrated by numerical experiments, making it suitable for solving the shipborne landing guidance problem in real time.

3.2.3. Outer-loop guidance implementation

The outer-loop guidance employs an iterative procedure, as formalized in Algorithm 1. The subscript k denotes the guidance cycle, i denotes the iteration index within a guidance cycle, and j denotes the mesh node index. Variables with two subscripts represent time-dependent functions, whereas variables with three subscripts denote the specific values of those functions at the j -th mesh node. Within each guidance cycle, the HSCP method iteratively refines landing trajectories until the convergence criterion set by ϵ_{tol} is satisfied or the iteration limit i_{max} is reached.

For numerical implementation, Problem 3 is transcribed into a second-order cone programming problem using trapezoidal discretization with N mesh points uniformly distributed over $\tau \in [0, 1]$. The dy-

Algorithm 1: Outer-loop guidance algorithm with HSCP.

Input: Current state $(\mathbf{r}_0, \mathbf{V}_0, \phi_0, m_0)$, predicted target

$$(\hat{\mathbf{r}}_{I, \text{deck}}^{[k]}, \hat{\mathbf{V}}_{I, \text{deck}}^{[k]})$$

Output: Optimized trajectory $(\mathbf{r}_I^{[k]}, \mathbf{V}_I^{[k]}, \phi^{[k]}, m^{[k]}, F^{[k]}, \omega^{[k]}, t_f^{[k]})$

```

1 Initialize  $(\theta_{k,0}, \lambda_{k,0})$  from previous guidance cycle solution;
2 Set iteration counter  $i \leftarrow 0$ ;
3 while true do
4   Construct Problem 3 using  $(\theta_{k,i}, \lambda_{k,i})$ ;
5   Solve Problem 3 to obtain  $(\theta_{k,i+1}, \lambda_{k,i+1})$ ;
6   if  $\max \|(\mathbf{r}_I^{\theta_{k,i+1}}, \mathbf{V}_I^{\theta_{k,i+1}}, \phi^{\theta_{k,i+1}}) - (\mathbf{r}_I^{\theta_{k,i}}, \mathbf{V}_I^{\theta_{k,i}}, \phi^{\theta_{k,i}})\|_{\infty} < \epsilon_{\text{tol}}$ 
7     then
8       break
9   else
10    if  $i + 1 < i_{\text{max}}$  then
11       $i \leftarrow i + 1$ 
12    else
13      break
14  end
15 end
16 return  $\theta_{k,i+1}$  as optimized trajectory.
```

namics constraint is enforced at collocation points through the trapezoidal rule:

$$(N-1)(\mathbf{x}_{k,i,j+1} - \mathbf{x}_{k,i,j}) = (\mathbf{x}'_{k,i,j+1} + \mathbf{x}'_{k,i,j})/2; \quad j = 1, \dots, N-1 \quad (15)$$

After solving the second-order cone programming problem, the costate $\lambda_{k,i+1}$ is reconstructed from the dual variables associated with the dynamics constraint.

3.3. Inner-loop guidance with IMBO and DOFE

The inner-loop guidance consists of three parts. First, an internal model-based observer is designed to estimate the deck motion. Then, a deck oscillation frequency estimator is developed, enabling prediction of future deck motion. Finally, a feedback guidance law is designed to track the outer-loop generated trajectory with compensation for unpredicted deck motion.

3.3.1. Trajectory tracking guidance problem

The basis of inner-loop guidance is the trajectory tracking error defined by

$$\mathbf{e}_r = \mathbf{r}_{\mathcal{E}} - \mathbf{r}_{\mathcal{E}}^{[k]} \quad (16)$$

where

$$\mathbf{r}_{\mathcal{E}} = \mathbf{r}_I - \mathbf{r}_{I, \text{deck}} \quad (17)$$

$$\mathbf{r}_{\mathcal{E}}^{[k]} = \mathbf{r}_I^{[k]} - \hat{\mathbf{r}}_{I, \text{deck}}^{[k]} \quad (18)$$

Since the outer loop ensures that $\mathbf{r}_{\mathcal{E}}^{[k]}(t_f^{[k]}) = \mathbf{0}$, stabilizing $\mathbf{e}_r$ to zero implies that $\mathbf{r}_I(t_f^{[k]}) = \mathbf{r}_{I, \text{deck}}(t_f^{[k]})$, fulfilling the shipborne landing requirements.

3.3.2. Internal-model based observer (IMBO)

The drone ship motion is observed from the dynamical system of $\mathbf{r}_{\mathcal{E}}$. Substituting (4) into (17) gives

$$\dot{\mathbf{r}}_{\mathcal{E}} = \mathbf{r}_I - \mathbf{A}_0 - (\mathbf{L} \otimes \mathbf{I}_3) \mathbf{w} \quad (19)$$

Let $\mathbf{V}_{\mathcal{E}}$ denote the time derivatives of $\mathbf{r}_{\mathcal{E}}$. They satisfy the following dynamical system:

$$\dot{\mathbf{r}}_{\mathcal{E}} = \mathbf{V}_{\mathcal{E}} \quad (20)$$

$$\dot{\mathbf{V}}_{\mathcal{E}} = \mathbf{g}_I + (F \mathbf{R}_{IB} \mathbf{n}_B + \mathbf{A}_I)/m - (\mathbf{L} \mathbf{S}^2 \otimes \mathbf{I}_3) \mathbf{w} \quad (21)$$

To estimate $\mathbf{w}$, the state describing the motion of the drone ship, an internal model-based observer is designed as follows:

$$\dot{\xi} = (\mathbf{D} \otimes \mathbf{I}_3)\xi - (\mathbf{D}\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}} + (\mathbf{G} \otimes \mathbf{I}_3)[\mathbf{g}_I + (\mathbf{F}\mathbf{R}_{IB}\mathbf{n}_B + \mathbf{A}_I)/m] \quad (22)$$

where $\xi \in \mathbb{R}^{6s}$ is the observer state, $\mathbf{D} \in \mathbb{R}^{2s \times 2s}$ and $\mathbf{G} \in \mathbb{R}^{2s \times 1}$ are the parameters of the observer, selected such that $\mathbf{D}$ is Hurwitz and $(\mathbf{D}, \mathbf{G})$ is controllable. A notable feature of the internal model-based observer (22) is that it does not depend on the unknown deck oscillation parameters $(\mathbf{S}, \mathbf{L}, \mathbf{w}_0, \mathbf{A}_0)$, yet it converges to $\mathbf{w}$ at an exponential rate, as shown subsequently.

The observer error is defined as follows:

$$\mathbf{e}_{\xi} = (\mathbf{M} \otimes \mathbf{I}_3)\mathbf{w} - \xi + (\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}} \quad (23)$$

where $\mathbf{M} \in \mathbb{R}^{2s \times 2s}$ is the unique solution to the Sylvester equation:

$$\mathbf{M}\mathbf{S} - \mathbf{D}\mathbf{M} = \mathbf{G}\mathbf{L}\mathbf{S}^2 \quad (24)$$

This Sylvester equation has a unique solution because the matrices $\mathbf{S}$ and $\mathbf{D}$ have no common eigenvalues: the eigenvalues of $\mathbf{S}$ have zero real parts, while the eigenvalues of $\mathbf{D}$, being a Hurwitz matrix, have strictly negative real parts. Moreover, according to [36], the unique solution $\mathbf{M}$ is nonsingular because $(\mathbf{D}, \mathbf{G})$ is controllable by design and $(\mathbf{S}, \mathbf{L}\mathbf{S}^2)$ is observable. The observability of $(\mathbf{S}, \mathbf{L}\mathbf{S}^2)$ is proven in Appendix A.

Consider the time derivative of the observer error $\mathbf{e}_{\xi}$:

$$\begin{aligned} \dot{\mathbf{e}}_{\xi} &= (\mathbf{M} \otimes \mathbf{I}_3)\dot{\mathbf{w}} - \dot{\xi} + (\mathbf{G} \otimes \mathbf{I}_3)\dot{\mathbf{V}}_{\mathcal{E}} \\ &= (\mathbf{M}\mathbf{S} \otimes \mathbf{I}_3)\mathbf{w} - (\mathbf{D} \otimes \mathbf{I}_3)\xi + (\mathbf{D}\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}} - (\mathbf{G}\mathbf{L}\mathbf{S}^2 \otimes \mathbf{I}_3)\mathbf{w} \end{aligned} \quad (25)$$

Substituting the Sylvester Eq. (24) into (25) gives

$$\begin{aligned} \dot{\mathbf{e}}_{\xi} &= (\mathbf{D}\mathbf{M} \otimes \mathbf{I}_3)\mathbf{w} - (\mathbf{D} \otimes \mathbf{I}_3)\xi + (\mathbf{D}\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}} \\ &= (\mathbf{D} \otimes \mathbf{I}_3)\mathbf{e}_{\xi} \end{aligned} \quad (26)$$

A closed-form solution to (26) is

$$\mathbf{e}_{\xi}(t) = e^{(\mathbf{D} \otimes \mathbf{I}_3)t} \mathbf{e}_{\xi}(0) \quad (27)$$

Because the matrix $\mathbf{D}$ is Hurwitz, all eigenvalues of $\mathbf{D} \otimes \mathbf{I}_3$ are strictly negative. Therefore, there exist positive constants c'_{ξ} and λ_{ξ} such that

$$\|e^{(\mathbf{D} \otimes \mathbf{I}_3)t}\|_2 \leq c'_{\xi} e^{-\lambda_{\xi} t} \quad (28)$$

Selecting $c_{\xi} = c'_{\xi} \|\mathbf{e}_{\xi}(0)\|_2$ establishes the exponential convergence of the internal model-based observer:

$$\|\mathbf{e}_{\xi}(t)\|_2 \leq c_{\xi} e^{-\lambda_{\xi} t} \quad (29)$$

According to (23), $(\mathbf{M}^{-1} \otimes \mathbf{I}_3)[\xi - (\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}}]$ converges exponentially to $\mathbf{w}$, providing the basis for predicting the deck motion.

3.3.3. Deck oscillation frequency estimator (DOFE)

For predicting $\mathbf{r}_{I,\text{deck}}$ and $\mathbf{V}_{I,\text{deck}}$, the deck oscillation frequency needs to be estimated. Define $\mathbf{w}_M = (\mathbf{M} \otimes \mathbf{I}_3)\mathbf{w}$. The dynamical system of $\mathbf{w}_M$ is given by

$$\begin{aligned} \dot{\mathbf{w}}_M &= [(\mathbf{M}\mathbf{S}\mathbf{M}^{-1}) \otimes \mathbf{I}_3]\mathbf{w}_M \\ &= [(\mathbf{D} + \mathbf{G}\mathbf{L}\mathbf{S}^2\mathbf{M}^{-1}) \otimes \mathbf{I}_3]\mathbf{w}_M \end{aligned} \quad (30)$$

Define constant row vectors $\mathbf{N}_1 = \mathbf{L}\mathbf{S}\mathbf{M}^{-1}$ and $\mathbf{N}_2 = \mathbf{L}\mathbf{S}^2\mathbf{M}^{-1}$. If $\mathbf{N}_1$ and $\mathbf{N}_2$ are known, the platform's velocity can be predicted from

$$\begin{aligned} \mathbf{V}_{I,\text{deck}}(t) &= (\mathbf{L}\mathbf{S} \otimes \mathbf{I}_3)\mathbf{w}_M(t) \\ &= (\mathbf{L}\mathbf{S}\mathbf{M}^{-1} \otimes \mathbf{I}_3)\mathbf{w}_M(t) \\ &= (\mathbf{N}_1 \otimes \mathbf{I}_3)e^{[(\mathbf{D} + \mathbf{G}\mathbf{N}_2) \otimes \mathbf{I}_3](t-t_k)}[\xi(t_k) - (\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}}(t_k) + \mathbf{e}_{\xi}(t_k)] \end{aligned} \quad (31)$$

by neglecting the unknown but exponentially convergent term $\mathbf{e}_{\xi}$. However, the row vectors $\mathbf{N}_1$ and $\mathbf{N}_2$ that encode the oscillation frequency are unknown. From the Sylvester equation in (24), it follows that:

$$\mathbf{N}_2 = \mathbf{N}_1\mathbf{D}/(1 - \mathbf{N}_1\mathbf{G}) \quad (32)$$

This relationship indicates that only $\mathbf{N}_1$ needs to be estimated, as $\mathbf{N}_2$ can be determined once $\mathbf{N}_1$ is known.

The problem of estimating $\mathbf{N}_1$ is reformulated as a linear regression problem. Note that $\mathbf{N}_1$ can be viewed as an output matrix:

$$\mathbf{V}_{I,\text{deck}}(t) = (\mathbf{N}_1 \otimes \mathbf{I}_3)[\xi(t) - (\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}}(t)] + (\mathbf{N}_1 \otimes \mathbf{I}_3)\mathbf{e}_{\xi}(t) \quad (33)$$

Integrating the left and right-hand sides of (33) over the time interval $[t_k, t_{k+1}]$ gives

$$\mathbf{x}_k \mathbf{N}_1^T = \mathbf{y}_k - \mathbf{z}_k \quad (34)$$

where $\mathbf{x}_k \in \mathbb{R}^{3 \times 2s}$, $\mathbf{y}_k \in \mathbb{R}^3$, and $\mathbf{z}_k \in \mathbb{R}^3$ are matrices and vectors such that

$$\text{vec}(\mathbf{x}_k) = \int_{t_k}^{t_{k+1}} \xi(t) dt - (\mathbf{G} \otimes \mathbf{I}_3)[\mathbf{r}_{\mathcal{E}}(t_{k+1}) - \mathbf{r}_{\mathcal{E}}(t_k)] \quad (35)$$

$$\mathbf{y}_k = \mathbf{r}_{I,\text{deck}}(t_{k+1}) - \mathbf{r}_{I,\text{deck}}(t_k) \quad (36)$$

$$\mathbf{z}_k = (\mathbf{N}_1 \otimes \mathbf{I}_3) \int_{t_k}^{t_{k+1}} \mathbf{e}_{\xi} dt \quad (37)$$

By aggregating (34) over the previous n guidance cycles, one formulates the following system of equations:

$$\begin{bmatrix} \mathbf{x}_{k-n} \\ \vdots \\ \mathbf{x}_{k-1} \end{bmatrix} \mathbf{N}_1^T = \begin{bmatrix} \mathbf{y}_{k-n} \\ \vdots \\ \mathbf{y}_{k-1} \end{bmatrix} - \begin{bmatrix} \mathbf{z}_{k-n} \\ \vdots \\ \mathbf{z}_{k-1} \end{bmatrix} \quad (38)$$

This can be written more compactly as a matrix equation:

$$\mathbf{X}_k \mathbf{N}_1^T = \mathbf{Y}_k - \mathbf{Z}_k \quad (39)$$

Treating $\mathbf{Z}_k$ as noise, the least squares solution to (39) provides an adaptation law to estimate $\mathbf{N}_1$ in the k -th guidance cycle:

$$\mathbf{N}_1^{[k]} = \mathbf{Y}_k^T \mathbf{X}_k (\mathbf{X}_k^T \mathbf{X}_k)^{-1} \quad (40)$$

The exponential convergence of $\mathbf{Z}_k$ demonstrates that $\mathbf{N}_1^{[k]}$ converges to $\mathbf{N}_1$ at an exponential rate. According to (32), $\mathbf{N}_2$ can then be estimated as

$$\mathbf{N}_2^{[k]} = \mathbf{N}_1^{[k]} \mathbf{D} / (1 - \mathbf{N}_1^{[k]} \mathbf{G}) \quad (41)$$

Now all necessary components for predicting the deck oscillation have been established. Consider the platform's velocity in (31). Dropping the unknown but convergent term $\mathbf{e}_{\xi}$ and replacing matrices $(\mathbf{N}_1, \mathbf{N}_2)$ with their estimates $(\hat{\mathbf{N}}_1^{[k]}, \hat{\mathbf{N}}_2^{[k]})$ gives the prediction of the platform's velocity:

$$\hat{\mathbf{V}}_{I,\text{deck}}^{[k]}(t) = (\hat{\mathbf{N}}_1^{[k]} \otimes \mathbf{I}_3) e^{[(\mathbf{D} + \mathbf{G}\hat{\mathbf{N}}_2^{[k]}) \otimes \mathbf{I}_3](t-t_k)}[\xi(t_k) - (\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}}(t_k)] \quad (42)$$

The platform's position can then be predicted from integration:

$$\hat{\mathbf{r}}_{I,\text{deck}}^{[k]}(t) = \mathbf{r}_{I,\text{deck}}(t_k) + \int_{t_k}^t \hat{\mathbf{V}}_{I,\text{deck}}^{[k]}(\tau) d\tau \quad (43)$$

An analytical solution to (43) exists if $\mathbf{D} + \mathbf{G}\hat{\mathbf{N}}_2^{[k]}$ is invertible:

$$\begin{aligned} \hat{\mathbf{r}}_{I,\text{deck}}^{[k]}(t) &= \mathbf{r}_{I,\text{deck}}(t_k) + (\hat{\mathbf{N}}_1^{[k]} \otimes \mathbf{I}_3) (e^{[(\mathbf{D} + \mathbf{G}\hat{\mathbf{N}}_2^{[k]}) \otimes \mathbf{I}_3](t-t_k)} - \mathbf{I}_{6s}) \\ &\quad [(\mathbf{D} + \mathbf{G}\hat{\mathbf{N}}_2^{[k]})^{-1} \otimes \mathbf{I}_3][\xi(t_k) - (\mathbf{G} \otimes \mathbf{I}_3)\mathbf{V}_{\mathcal{E}}(t_k)] \end{aligned} \quad (44)$$

where $\mathbf{I}_{6s}$ is the unit matrix of dimension $6s$.

3.3.4. Feedback guidance implementation

In the k -th guidance cycle, the reference trajectory comprises state history $(\mathbf{r}_I^{[k]}, \mathbf{V}_I^{[k]}, \phi^{[k]}, m^{[k]})$ and control history $(\mathbf{F}^{[k]}, \omega^{[k]})$. The dynamical system of the reusable rocket can be approximated around the reference trajectory using first-order Taylor expansion:

$$\delta \dot{\mathbf{r}}_I = \delta \mathbf{V}_I \quad (45)$$

$$\delta \dot{\mathbf{V}}_I = \mathbf{Q}_r^{[k]} \delta \mathbf{r}_I + \mathbf{Q}_V^{[k]} \delta \mathbf{V}_I + \mathbf{Q}_{\phi}^{[k]} \delta \phi + \mathbf{Q}_m^{[k]} \delta m + \mathbf{Q}_F^{[k]} \delta \mathbf{F} \quad (46)$$

$$\delta \dot{\phi} = \delta \omega \quad (47)$$

$$\delta \dot{m} = -\delta \mathbf{F} / V_{\text{ex}} \quad (48)$$

where $\delta \mathbf{r}_I$, $\delta \mathbf{V}_I$, $\delta \boldsymbol{\phi}$, δm , δF , and $\delta \boldsymbol{\omega}$ are the deviations from the reference trajectory. $\mathbf{Q}_r^{[k]}$, $\mathbf{Q}_V^{[k]}$, $\mathbf{Q}_\phi^{[k]}$, $\mathbf{Q}_m^{[k]}$, and $\mathbf{Q}_F^{[k]}$ are the Jacobian matrices of dynamics evaluated on the reference trajectory.

Using the approximation around the reference trajectory, the tracking error $\mathbf{e}_r$ and its time derivative $\dot{\mathbf{e}}_r$ can be rewritten as

$$\begin{aligned} \mathbf{e}_r &= (\mathbf{r}_I - \mathbf{r}_{I,\text{deck}}) - (\mathbf{r}_I^{[k]} - \mathbf{r}_{I,\text{deck}}^{[k]}) \\ &= \delta \mathbf{r}_I + (\mathbf{r}_{I,\text{deck}}^{[k]} - \mathbf{r}_{I,\text{deck}}) \end{aligned} \quad (49)$$

and

$$\dot{\mathbf{e}}_r = \delta \mathbf{V}_I + (\dot{\mathbf{r}}_{I,\text{deck}}^{[k]} - \dot{\mathbf{r}}_{I,\text{deck}}) \quad (50)$$

They satisfy the following dynamical system:

$$\begin{bmatrix} \dot{\mathbf{e}}_r \\ \dot{\mathbf{e}}_V \end{bmatrix} = \begin{bmatrix} \mathbf{Q}_r^{[k]} \delta \mathbf{r}_I + \mathbf{Q}_V^{[k]} \delta \mathbf{V}_I + \mathbf{Q}_m^{[k]} \delta m + \delta \mathbf{u} + \dot{\mathbf{r}}_{I,\text{deck}}^{[k]} - \dot{\mathbf{r}}_{I,\text{deck}} \\ \mathbf{e}_V \end{bmatrix} \quad (51)$$

where $\delta \mathbf{u}$ is the virtual control defined by

$$\delta \mathbf{u} = \mathbf{Q}_\phi^{[k]} \delta \boldsymbol{\phi} + \mathbf{Q}_F^{[k]} \delta F \quad (52)$$

To stabilize the tracking error, the inner-loop guidance employs the following feedback guidance law:

$$\begin{aligned} \delta \mathbf{u} &= (\mathbf{K}_r - \mathbf{Q}_r^{[k]}) \delta \mathbf{r}_I + (\mathbf{K}_V - \mathbf{Q}_V^{[k]}) \delta \mathbf{V}_I - \mathbf{Q}_m^{[k]} \delta m \\ &\quad - \mathbf{K}_r (\mathbf{r}_{I,\text{deck}} - \mathbf{r}_{I,\text{deck}}^{[k]}) - \mathbf{K}_V (\mathbf{V}_{I,\text{deck}} - \dot{\mathbf{r}}_{I,\text{deck}}^{[k]}) \end{aligned} \quad (53)$$

where $\mathbf{K}_r$ and $\mathbf{K}_V$ are constant feedback gain matrices. The first three terms in (53) provide state feedback compensation to track the reference trajectory, while the latter term compensates for unpredicted deck motion by feeding back the error between predictions and real-time measurements.

Substituting the trajectory tracking law (53) into the error dynamics (51) yields the following closed-loop system:

$$\begin{bmatrix} \dot{\mathbf{e}}_r \\ \dot{\mathbf{e}}_V \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_3 \\ \mathbf{K}_r & \mathbf{K}_V \end{bmatrix} \begin{bmatrix} \mathbf{e}_r \\ \mathbf{e}_V \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \dot{\mathbf{r}}_{I,\text{deck}}^{[k]} - \dot{\mathbf{r}}_{I,\text{deck}} \end{bmatrix} \quad (54)$$

System (54) represents a linear time-invariant system driven by the velocity prediction error rate. As the prediction error rate converges to zero, the tracking error ($\mathbf{e}_r$, $\mathbf{e}_V$) converges to zero at an exponential rate if $\mathbf{K}_r$ and $\mathbf{K}_V$ are selected such that the system matrix is Hurwitz.

The virtual control $\delta \mathbf{u}$ is allocated to the actual control variables $\delta \boldsymbol{\phi}_{\text{cmd}}$ and δF_{cmd} through the inverse of the control effectiveness matrix:

$$\begin{bmatrix} \delta \boldsymbol{\phi}_{\text{cmd}} \\ \delta F_{\text{cmd}} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}_\phi^{[k]} & \mathbf{Q}_F^{[k]} \end{bmatrix}^{-1} \delta \mathbf{u} \quad (55)$$

The attitude deviation $\delta \boldsymbol{\phi}_{\text{cmd}}$ is regulated through a proportional feedback controller on the attitude rate:

$$\delta \boldsymbol{\omega}_{\text{cmd}} = \mathbf{K}_\phi (\delta \boldsymbol{\phi}_{\text{cmd}} - \delta \boldsymbol{\phi}) \quad (56)$$

where $\mathbf{K}_\phi$ is a positive definite gain matrix. Finally, the guidance commands are obtained by superimposing the outer-loop provided reference trajectory with the inner-loop provided feedback corrections:

$$\begin{bmatrix} F_{\text{cmd}} \\ \boldsymbol{\omega}_{\text{cmd}} \end{bmatrix} = \begin{bmatrix} F^{[k]} \\ \boldsymbol{\omega}^{[k]} \end{bmatrix} + \begin{bmatrix} \delta F_{\text{cmd}} \\ \delta \boldsymbol{\omega}_{\text{cmd}} \end{bmatrix} \quad (57)$$

3.4. Guidance implementation

The proposed DR-COBG comprises four interconnected components: trajectory optimization based on HSCP, an IMBO, a DOFE, and a trajectory tracking controller. To balance computational efficiency with performance, these components operate at different frequencies in two parallel processes: a long-duration guidance process and a short-duration guidance process. The long-duration process updates the drone ship motion prediction and the landing trajectory at a lower frequency, while the short-duration process executes the trajectory tracking controller at a higher frequency to ensure rapid feedback. The complete implementation is detailed in Algorithm 2. Moreover, several numerical methods are used in the outer-loop guidance. The Hessian of f is evaluated using the central difference method. The second-order cone programming problems are solved using ECOS [37].

Algorithm 2: DR-COBG implementation.

Input: Current flight state, drone ship measurements
Output: Control commands to actuators

- 1 **Long-duration guidance process (low frequency):**
- 2 DOFE: Update $N_1^{[k]}$, $N_2^{[k]}$ from measurements $\mathbf{x}_{k-1}$, $\mathbf{y}_{k-1}$;
- 3 DOFE: Predict drone ship state ($\hat{\mathbf{r}}_{I,\text{deck}}^{[k]}$, $\dot{\hat{\mathbf{r}}}_{I,\text{deck}}^{[k]}$);
- 4 **if** $t < t_{\text{predict}}$ **then**
- 5 | Set target $\leftarrow$ current drone ship state;
- 6 **else**
- 7 | Set target $\leftarrow$ predicted drone ship state;
- 8 **end**
- 9 **if** $t_f^{[k-1]} - t > t_{\text{stop}}$ **then**
- 10 | HSCP: Update landing trajectory using target;
- 11 **end**
- 12 Store updated trajectory for short-duration process.
- 13 **Short-duration guidance process (high frequency):**
- 14 IMBO and DOFE: Integrate using Euler's method;
- 15 Interpolate reference from stored trajectory;
- 16 Tracking controller: Compute control commands from tracking errors;
- 17 Send control commands (F_{cmd} , $\boldsymbol{\omega}_{\text{cmd}}$) to actuators.

Table 1

Landing condition and reusable rocket data.

Parameter	Value
Gravitational acceleration $\mathbf{g}_I$	(0, -9.81, 0) m/s ²
Thrust direction $\mathbf{n}_B$	(1, 0, 0)
Number of sinusoids s	2
Mean deck position $\mathbf{A}_0$	(0, 0, 0) m
First oscillation amplitude $\mathbf{A}_1$	(4.0736, 0.1349, 2.5290) m
First oscillation frequency ω_1	0.6 rad/s
First oscillation phase ϕ_1	0.1576 rad
Second oscillation amplitude $\mathbf{A}_2$	(1.0938, 0.0951, 0.5570) m
Second oscillation frequency ω_2	1.5 rad/s
Second oscillation phase ϕ_2	0.9575 rad
Control penalties (k_F , k_ω)	(1×10^{-12} , 25)
Initial time t_0	0 s
Initial position $\mathbf{r}_0$	(-300, 1000, 100) m
Initial velocity $\mathbf{V}_0$	(20, -50, 0) m/s
Initial attitude $\boldsymbol{\phi}_0$	(0, 90) deg
Initial mass m_0	85000 kg
Minimum thrust $F_{\min}$	500 kN
Maximum thrust $F_{\max}$	1500 kN
Exhaust velocity V_{ex}	3000 m/s
Reference area S_{ref}	16 m
Aerodynamic coefficient C_B	Reference [38]

4. Numerical studies

Numerical experiments are performed to validate the performance of the proposed DR-COBG. The landing condition and reusable rocket data are described in Table 1.

The parameters of DR-COBG are given as follows. The long-duration and short-duration guidance cycles are 0.05 s and 0.005 s, respectively. For the outer loop, the convergence criterion ϵ_{tol} is set to

$$\epsilon_{\text{tol}} = (1, 1, 1, 1, 1, 1, 0.01, 0.01, 0.1) \quad (58)$$

The trust-region radius κ is set to

$$\kappa = (1000, 1000, 1000, 100, 100, 100, 1, 1, 10000, 100, 500 \times 10^3, 10, 10, 100, 5) \quad (59)$$

The iteration limit $i_{\max}$ is 10. The number of mesh points N is 20. In the first guidance cycle, the initial guess is set as a linear interpolation from current state to the drone ship state. The initial guess for the costate is set to zero. The outer loop trajectory update stops as the time-to-go

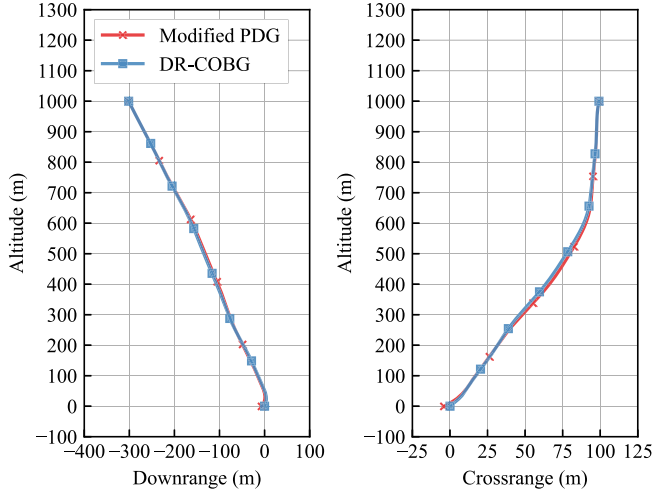


Fig. 4. Rocket trajectory (relative to drone ship).

becomes lower than $t_{\text{stop}} = 5$ s. Computation times were measured on a commercial laptop equipped with an AMD Ryzen 7 7840H processor. For the inner loop, the IMBO is driven by the measured relative velocity and acceleration. Its coefficients D and G are set as follows:

$$D = \begin{bmatrix} -20 & 1 & 0 & 0 \\ 0 & -20 & 1 & 0 \\ 0 & 0 & -20 & 1 \\ 0 & 0 & 0 & -20 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (60)$$

The DOFE's coefficient n is set to 60. t_{predict} is set to 5 s. The feedback gains are set as follows:

$$K_r = \text{diag}(-1, -1, -1), \quad K_v = \text{diag}(-2, -2, -2), \quad K_\phi = \text{diag}(1, 1, 1) \quad (61)$$

The thrust and attitude control systems are formulated as first-order dynamic systems and the time constants are set to 0.1 s.

4.1. Comparison with a modified PDG method

Since no existing PDG methods have addressed the shipborne landing problem, this paper, for the sake of comparison, adapts the PDG method from reference [18]. The successive convexification framework from this reference is employed to solve Problem 1, updating the guidance target to the drone ship in each guidance cycle. The comparison re-

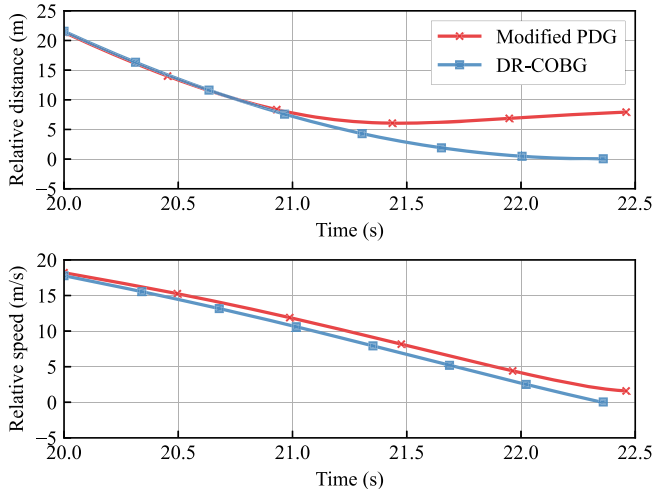


Fig. 5. Relative distance and speed history.

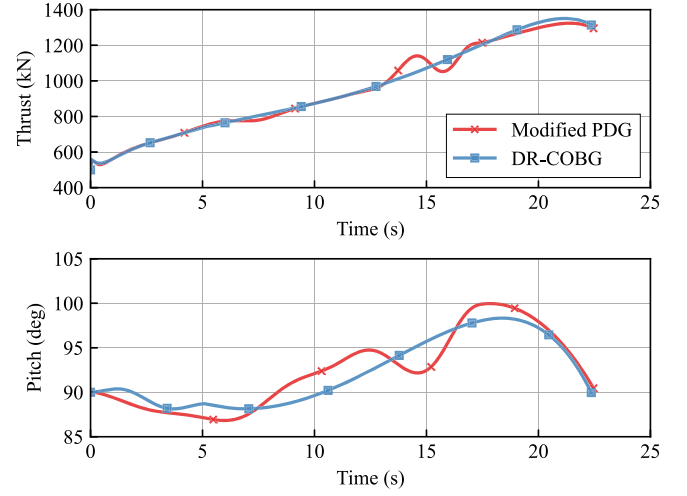


Fig. 6. Thrust and pitch angle history.

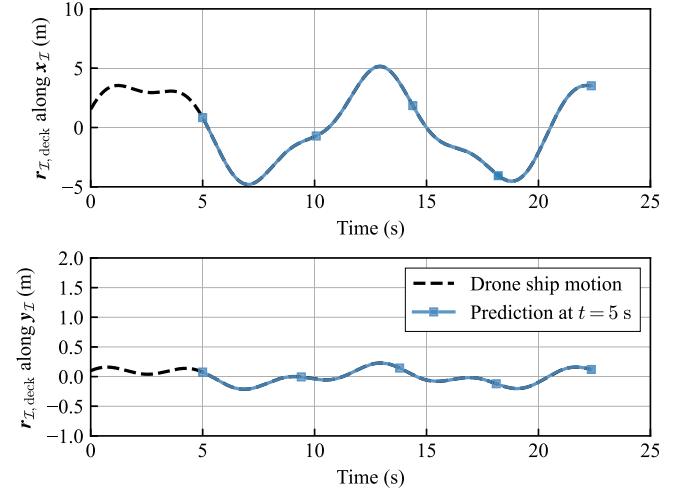


Fig. 7. Drone ship motion and DR-COBG's prediction.

sults with modified PDG and the proposed DR-COBG are given in Figs. 4 to 7 and Table 2.

The simulation results demonstrate the superiority of the proposed DR-COBG method for drone ship landing. Specifically, DR-COBG achieves a final position error of 0.0587 m, a significant improvement over the 7.9191 m error of the modified PDG. Velocity and attitude errors are also markedly reduced. Furthermore, the proposed method exhibits superior propellant efficiency and reduced control oscillation, as shown in Fig. 6. These improvements are attributable to the accurate prediction of the drone ship's motion by the IMBO and DOFE modules. As illustrated in Fig. 7, the prediction at $t = 5$ s aligns closely with the actual drone ship motion. This allows the DR-COBG to target a predicted deck state directly, rather than reacting to the platform's oscillations. The developed Hessian-enhanced SCP method also shows higher computational efficiency. For the shipborne landing guidance problem, the aver-

Table 2

Key performance metrics for drone ship landing.

Method	Position error	Velocity error (m/s)	Attitude error (deg)	Propellant usage (kg)
Modified PDG	7.9191 m	1.5709 m/s	0.5734 deg	7101.7kg
Proposed DR-COBG	0.0587 m	0.0764 m/s	0.0642 deg	7075.3kg

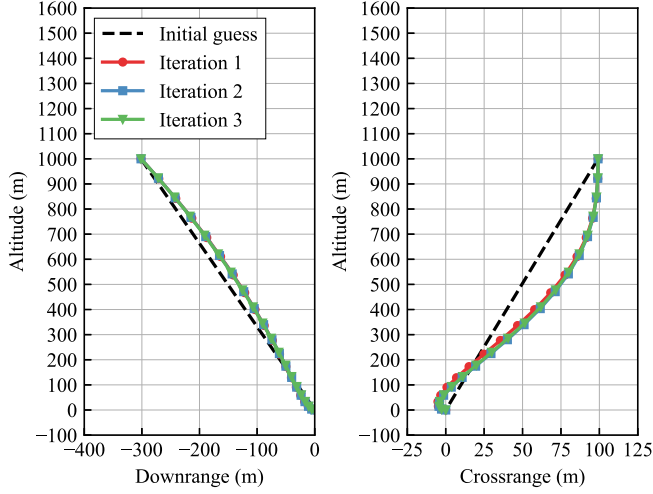
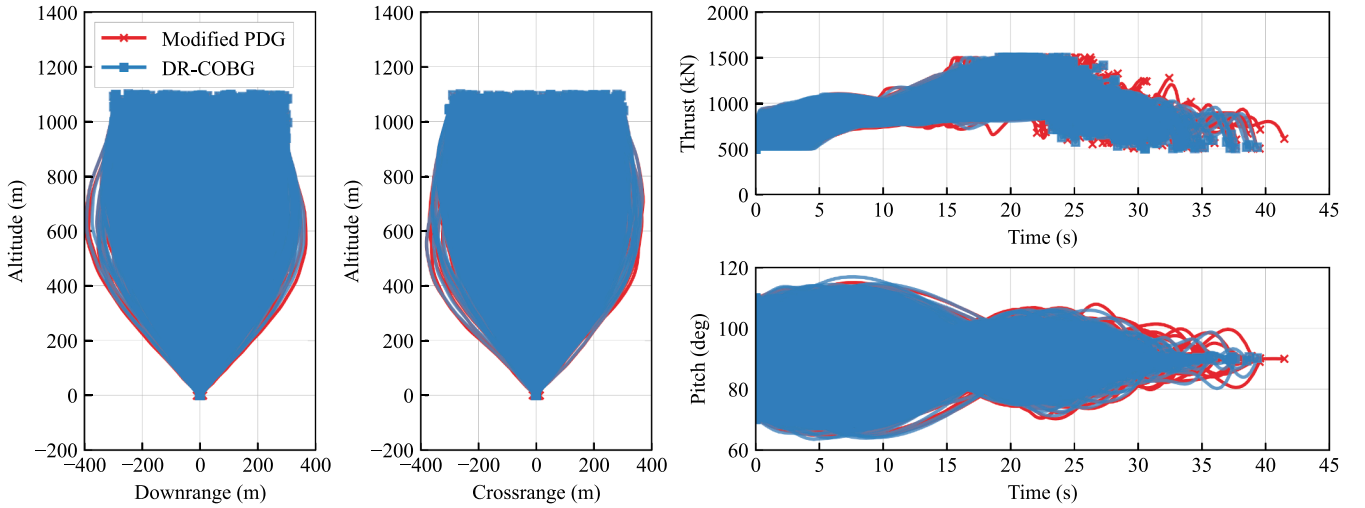


Fig. 8. Iteration history of rocket trajectory.

age solve time per guidance cycle of the proposed DR-COBG is 0.0065 s, a 39.3% reduction compared with modified PDG at 0.0107 s. The average number of iterations for DR-COBG and PDG are 1.0546 and 2.0457, respectively, showing that the proposed DR-COBG converged mostly in a single iteration. From the initial guess (a linear interpolation from current state to the drone ship state), the convergence process is shown in Fig. 8. Consequently, the DR-COBG delivers not only high landing accuracy but also lower propellant consumption, making it suitable for autonomous rocket landing on moving drone ships.

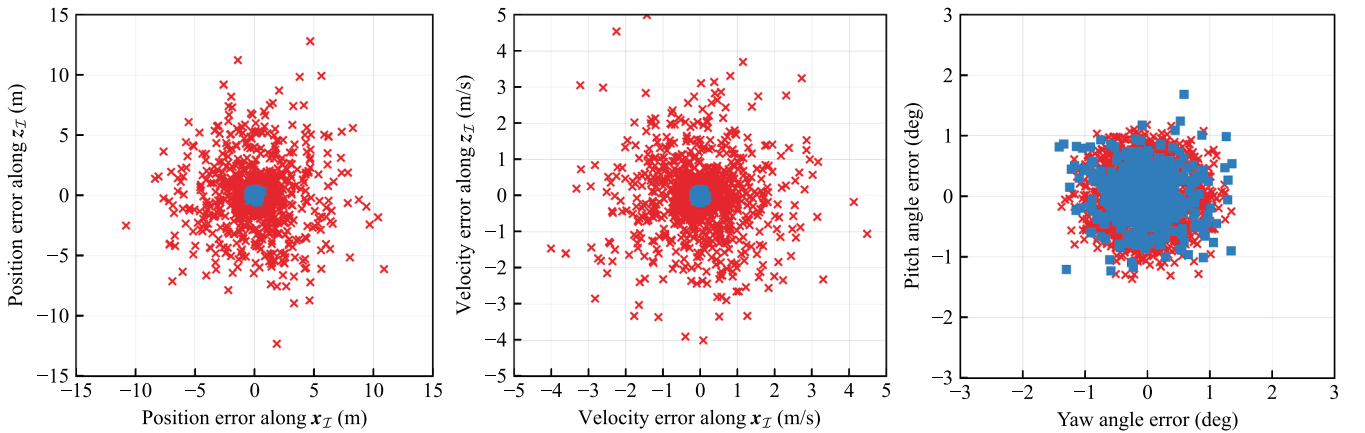
4.2. Monte Carlo experiments under dispersions

To further evaluate the proposed DR-COBG method, Monte Carlo experiments are conducted under various dispersions to demonstrate the adaptability of DR-COBG across different scenarios. Specifically, the initial position is uniformly distributed between $[-300, 300]$ m for the x_I and z_I directions, and $[900, 1100]$ m for the y_I direction. The initial velocity is uniformly distributed between $[-20, 20]$ m/s for the x_I and z_I directions, and $[-55, -45]$ m/s for the y_I direction. The initial attitude is uniformly distributed between $[-20, 20]$ deg. The drone ship oscillation



(a) Rocket trajectory (relative to drone ship)

(b) Thrust and pitch angle history



(c) Error in landing position

(d) Error in landing velocity

(e) Error in landing attitude

Fig. 9. Monte Carlo results.

Table 3
Statistical distribution of performance metrics across Monte Carlo experiments.

Method	Measure	Position error (m)			Velocity error (m/s)			Attitude error (deg)		Propellant usage (kg)
		x_I	y_I	z_I	x_I	y_I	z_I	Yaw	Pitch	
Modified PDG	Mean	0.0915	-0.4247	-0.0051	-0.0024	-0.0523	-0.0397	0.0058	0.0117	7995.7635
	Std	2.5004	0.5860	2.6583	0.9923	0.2838	1.0291	0.5687	0.5770	912.3881
Proposed DR-COBG	Mean	0.0048	0.0490	-0.0029	0.0013	0.0833	-0.0009	-0.0084	0.0123	7958.9179
	Std	0.1306	0.0446	0.1410	0.0444	0.0435	0.0463	0.3287	0.2981	875.2526

frequency is uniformly distributed in $[0.3, 0.6]$ and $[1.5, 2.0]$ rad/s. The atmospheric density is uniformly distributed within $\pm 20\%$ of the nominal value. The wind velocity is uniformly distributed in $[0, 20]$ m/s, with the wind direction randomly selected. Moreover, estimation of the deck oscillation by the IMBO and DOFE is incorporated into the modified PDG to ensure a fair comparison. A total of 1000 Monte Carlo simulations are performed. The results are presented in Fig. 9 and Table 3.

The Monte Carlo simulations demonstrate the robust performance of the proposed DR-COBG method across a wide range of operating conditions. As shown in Fig. 9a, all 1000 trajectories successfully converge to the oscillating drone ship deck despite significant variations in initial conditions and deck oscillation parameters. The position errors at landing, displayed in Fig. 9c, maintain a tight distribution with standard deviations below 0.14 m in horizontal and 0.05 m in vertical axes. The velocity error's standard deviations are below 0.05 m/s in all three axes. Statistical analysis in Table 3 quantifies the performance improvements over the modified PDG. Even though the modified PDG is incorporated with prediction of ship motion, the DR-COBG method still maintains better landing accuracy. Additionally, when compared to the reported landing accuracy in reference [34], which achieved a position standard deviation of 0.04 m and velocity standard deviation of 0.14 m/s with no wind disturbance, these results underscore DR-COBG's effectiveness in shipborne landing problems: The DR-COBG approach maintains comparable landing precision under significant drone ship oscillations and wind disturbances, conditions not addressed in reference [34].

5. Conclusion

A disturbance rejection and convex optimization-based guidance law is developed for reusable rockets landing on oscillating drone ships. A dual-loop framework is proposed, where the outer loop generates optimal trajectories toward predicted deck states using a novel Hessian-enhanced sequential convex programming method, while the inner loop tracks these trajectories using an internal model-based observer and deck oscillation frequency estimator.

Numerical studies validate the proposed guidance law, demonstrating precise touchdown with sub-meter position accuracy and robust performance across various dispersions. The method achieves both high landing accuracy and propellant efficiency by accurately predicting and compensating for deck oscillations, effectively bridging the gap between conventional rocket guidance and moving platform landing.

Despite these advantages, the current framework relies on the identification of sinusoidal oscillation frequencies. Consequently, handling complex non-sinusoidal ship motions associated with irregular sea states remains a limitation. Future work will focus on addressing non-sinusoidal motions. Additionally, an adaptive weighting scheme based on frequency estimation uncertainty will be developed to further enhance tracking robustness during shipborne landing. Moreover, while the proposed method demonstrated 100% convergence in Monte Carlo studies, a formal theoretical analysis of the HSCP convergence properties remains an open problem for future theoretical research.

CRedit authorship contribution statement

Jiawei Wang: Writing – original draft, Software, Methodology, Funding acquisition; **Ran Zhang:** Writing – review & editing, Method-

ology, Funding acquisition; **Huifeng Li:** Writing – review & editing, Methodology, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ran Zhang reports financial support was provided by National Natural Science Foundation of China and Beijing Natural Science Foundation. Huifeng Li reports financial support was provided by National Natural Science Foundation of China. Jiawei Wang reports financial support was provided by China Postdoctoral Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Observability of (S, LS^2)

The observability matrix of (S, LS^2) , denoted by $\mathcal{O} \in \mathbb{R}^{2s \times 2s}$, is given by

$$\mathcal{O} = \begin{bmatrix} (LS^2)^T & \dots & (LS^{2s+1})^T \end{bmatrix}^T = \begin{bmatrix} 0 & -\omega_1^2 & \dots & \dots & 0 & -\omega_s^2 \\ -\omega_1^3 & 0 & \dots & \dots & -\omega_s^3 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & (-1)^s \omega_1^{2s} & \dots & \dots & 0 & (-1)^s \omega_s^{2s} \\ (-1)^s \omega_1^{2s+1} & 0 & \dots & \dots & (-1)^s \omega_s^{2s+1} & 0 \end{bmatrix} \quad (\text{A.1})$$

which can be rearranged using a permutation matrix $\mathcal{Q}$ such that

$$\mathcal{O} = \mathcal{Q} \begin{bmatrix} \mathbf{0} & \mathcal{O}_A \\ \mathcal{O}_B & \mathbf{0} \end{bmatrix} \mathcal{Q}^T \quad (\text{A.2})$$

where $\mathcal{O}_A \in \mathbb{R}^{s \times s}$ and $\mathcal{O}_B \in \mathbb{R}^{s \times s}$ are given by

$$\mathcal{O}_A = \begin{bmatrix} -\omega_1^3 & \dots & -\omega_s^3 \\ \vdots & \vdots & \vdots \\ (-1)^s \omega_1^{2s+1} & \dots & (-1)^s \omega_s^{2s+1} \end{bmatrix} \quad (\text{A.3})$$

$$\mathcal{O}_B = \begin{bmatrix} -\omega_1^2 & \dots & -\omega_s^2 \\ \vdots & \vdots & \vdots \\ (-1)^s \omega_1^{2s} & \dots & (-1)^s \omega_s^{2s} \end{bmatrix} \quad (\text{A.4})$$

Both matrices can be factored as $\mathcal{O}_A = \mathcal{D}_A \mathcal{V}_A$ and $\mathcal{O}_B = \mathcal{D}_B \mathcal{V}_B$, where $\mathcal{D}_A$ and $\mathcal{D}_B$ are invertible diagonal matrices containing the alternating

signs, while $\mathcal{V}_A$ and $\mathcal{V}_B$ are submatrices of the Vandermonde matrix associated with $(\omega_1, \dots, \omega_s)$. Because the frequencies $\omega_1, \dots, \omega_s$ are distinct, the matrices $\mathcal{O}_A, \mathcal{O}_B$, and thereby the observability matrix $\mathcal{O}$ have full rank. Hence, (S, LS^2) is observable.

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